QR Code for
report verification

Case ID	:	2600132859	Sample Type	:	BLOOD IN CCF DNA TUBE
Name	:	MR. RAJENDRA PRASAD (2479921) (OQG2604220384)	Date & Time Collected	:	22-Apr-2026 12:00 AM
Sex/Age	:	Male/72 Years	Date & Time Received	:	23-Apr-2026 01:49 PM
Bill. Loc.	:	OncQuest Laboratories Ltd., New Delhi	Date & Time Reported	:	11-May-2026 06:33 PM
Ref. By	:		Report Version	:	1
Indication	:				

Liquidseq Comprehensive Genomic Profile (CGP) Panel On Illumina Novaseq 6000 Platform

Clinical Indication:

Features are consistent with squamous cell carcinoma

Negative

(No clinically significant Variants detected)

Key Findings:

Tier	Gene & Transcript	Variant	Exon	Coverage/VAF
		-Negative-		

Fusion Genes	Breakpoints	Total Reads
	-Negative-	

CNV Type	Genes	Copy Number
	-Negative-	

Biomarkers Results:

Assay Biomarker	Result	Interpretation
Microsatellite Instability (MSI)	Stable	Stable Microsatellite detected
Tumor Mutation Burden (TMB)	5.45 (Low)	Unlikely to benefit from Immuno-Oncology therapy

Test Description

Liquidseq Comprehensive Genomic Profile (CGP) Panel is advancing precision oncology research through the analysis

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of multiple relevant biomarkers in a single next-generation sequencing (NGS) test.

The Liquidseq Comprehensive Genomic Profile (CGP) Panel assay analyses **543** unique genes (**537** DNA + **6** Fusions driver genes) for single gene and multiple gene biomarker insights, detects single gene biomarkers from all variant types including **SNVs, indels, CNVs, known and novel fusions, and splice variants.** Analyze complex multi-gene biomarkers for mutational signatures, including microsatellite instability (**MSI**) & Tumor Mutation burden (**TMB**) for immunotherapy research.

The performance of the panel was verified using both commercially sourced reference controls and cell-free DNA (cfTNA)samples.

Test Methodology

Extraction & Library Preparation: Cell free Total Nucleic acid (cfTNA) extracted from the liquid biopsy sample is used for targeted gene capture using a custom designed hybrid capture panel. In process quality controls were determined for prepared library.

Sequencing: The libraries were sequenced at mean depth: **>25,000x** on **Illumina** Novaseq- next generation sequencing platform.

Data Analysis: Sequencing data are analyzed using a customized and validated bioinformatics pipeline capable of detecting various classes of genomic alterations, including single nucleotide variants (**SNVs**), **Indels**, copy number variations (**CNVs**) and **fusions**. As part of the streamlined bioinformatics workflow, Multi-gene Biomarkers such as Microsatellite Instability (**MSI**) and Tumor Mutation Burden (**TMB**) Score is calculated. Reportable genomic alterations and fusions are prioritized, classified, and reported based on AMP-ASCO-CAP guidelines and NCCN guidelines.

Limit of Detection (LOD): The LOD for SNVs and Short InDels is 0.1% Variant allele Frequency (VAF) and for Fusions is 05 supporting reads.

Notes

Treatment decisions should not be based solely on the results of this test. Findings must be interpreted in the context of the patient's complete clinical profile—including pathological, radiological, and family history—to guide diagnosis, prognosis, and therapeutic decisions.

Therapy-related information in this report is derived from multiple sources including FDA-approved drug data, NCCN guidelines, peer-reviewed literature, standard clinical databases, and biomarker evidence strength. However, therapy options listed may not be suitable or beneficial for all patients. This report aims to provide a summary of potentially effective treatments, therapies that may pose increased risks, and those that may have limited benefit, based on genomic alterations identified. Prognostic and diagnostic biomarkers relevant to the disease context may also be reported.

SN Genelab Pvt Ltd does not guarantee completeness, accuracy, or eligibility for trial enrollment. Physicians must verify eligibility criteria for specific trials independently.

Identification of a genomic alteration does not guarantee therapeutic efficacy or predict ineffectiveness. Final decisions on treatment should rest with the treating physician. The absence of reported therapeutic options does not imply ineffectiveness or lack of safety of any medication selected independently.

All variant classifications and related clinical information are based on the latest evidence from peer-reviewed publications, public clinical databases, and medical guidelines (e.g., NCCN, ASCO, AMP) available at the time of report generation. However, literature is continuously evolving, and classification may change with emerging evidence.

The assay is designed and validated for somatic variant detection, it does not differentiate between somatic and germline variants. If indicated, germline testing and genetic counselling are recommended.

Poor-quality cfDNA may fail QC thresholds at various stages such as cfDNA extraction, library preparation, or bioinformatics, and may result in assay failure or compromised results, such as low gene coverage or shallow variant depth.

Variants with lower allele frequencies may be reported if they are of strong or potential clinical relevance, or if previously detected in the same patient. False positives or negatives below assay sensitivity cannot be ruled out.

CNVs/gene amplifications reported, Confirmation using orthogonal methods such as DDPCR is recommended when clinically indicated.

Variants in non-coding regions or deep intronic sequences are not assessed in this assay.

This assay is under the scope of NABL as Liquidseq Comprehensive Genomic Profiling Panel.

DNA Gene List				
A1CF	ABCB1	ABL1	ABL2	ABRAXAS1
ACSM2B	ACVR1	ACVR1B	ACVR2A	ADAM18
ADAMTS12	ADAMTS2	AKT1	AKT2	AKT3

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ALK	AMER1	ANO4	APC	AR
ARAF	ARHGAP35	ARID1A	ARID1B	ARID2
ARID5B	ARMC4	ASXL1	ASXL2	ATM
ATP1A1	ATR	ATRX	AURKA	AURKB
AURKC	AXIN1	AXIN2	AXL	B2M
BAP1	BARD1	BCL2	BCL2L12	BCL6
BCOR	BCORL1	BCR	BLM	BMP5
BMPR2	BRAF	BRCA1	BRCA2	BRINP3
BRIP1	BTK	C6	C8A	C8B
CACNA1D	CALR	CANX	CARD11	CASP8
CASR	CBFB	CBL	CCND1	CCND2
CCND3	CCNE1	CD163	CD274	CD276
CD79B	CDC73	CDH1	CDH10	CDK12
CDK2	CDK4	CDK6	CDKN1A	CDKN1B
CDKN2A	CDKN2B	CDKN2C	CHD4	CHEK1
CHEK2	CIC	CIITA	CNTN6	CNTNAP4
CNTNAP5	COL11A1	CREBBP	CSF1R	CSMD3
CTCF	CTLA4	CTNNB1	CTNND2	CUL1
CUL3	CUL4A	CUL4B	CYLD	CYP2C9
CYP2D6	CYSLTR2	DAXX	DCAF4L2	DCDC1
DDR1	DDR2	DDX3X	DGCR8	DICER1
DNMT3A	DOCK3	DPYD	DROSHA	DSC1
DSC3	E2F1	EED	EGFR	EIF1AX
ELF3	EMSY	ENO1	EP300	EPAS1
EPCAM	EPHA2	ERAP1	ERAP2	ERBB2
ERBB3	ERBB4	ERCC2	ERCC4	ERCC5
ERG	ERRFI1	ESR1	ETV1	ETV4
ETV5	ETV6	EZH1	EZH2	FAM135B
FANCA	FANCC	FANCD2	FANCE	FANCF
FANCG	FANCI	FANCL	FANCM	FARSB
FAS	FAT1	FAT3	FAT4	FBXW7
FGF19	FGF23	FGF3	FGF4	FGF7
FGF9	FGFR1	FGFR2	FGFR3	FGFR4
FLT3	FLT4	FOXA1	FOXL2	FOXO1

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FUBP1	FYN	GALNT17	GATA2	GATA3
GLI1	GLI3	GNA11	GNA13	GNAQ
GNAS	GPR158	GPS2	GRID2	H1-4
H2BC5	H3-3A	H3-3B	H3C2	H3F3A
HCN1	HDAC2	HDAC9	HIF1A	HIST1H3B
HIST1H3C	HLA-A	HLA-B	HLA-C	HNF1A
HRAS	ID3	IDH1	IDH2	IGF1R
IKBKB	IL6ST	IL7R	INPP4B	IRF4
IRS4	JAK1	JAK2	JAK3	KCND2
KCNH7	KCNJ5	KDM5C	KDM6A	KDR
KEAP1	KEL	KIR3DL1	KIT	KLF4
KLF5	KLHL13	KMT2A	KMT2B	KMT2C
KMT2D	KNSTRN	KRAS	KRTAP21-1	KRTAP6-2
LARP4B	LATS1	LATS2	LRP1B	LRRC7
MAGOH	MAP2K1	MAP2K2	MAP2K4	MAP2K7
MAP3K1	MAP3K4	MAP3K8	MAPK1	MAPK3
MAPK8	MARCO	MAX	MCL1	MDM2
MDM4	MECOM	MED12	MEF2B	MEN1
MET	MGA	MITF	MLH1	MLH3
MPL	MRE11	MSH2	MSH3	MSH6
MTAP	MTOR	MTUS2	MUC16	MUTYH
MYB	MYBL1	MYC	MYCL	MYCN
MYD88	MYOD1	NBN	NCOR1	NF1
NF2	NFE2L2	NLRC5	NOL4	NOTCH1
NOTCH2	NOTCH3	NOTCH4	NPM1	NRAS
NRXN1	NSD2	NT5C2	NTRK1	NTRK2
NTRK3	NUP93	NUTM1	NYAP2	OR10G8
OR2G6	OR2L13	OR2L2	OR2L8	OR2M3
OR2T3	OR2T33	OR2T4	OR2W3	OR4A15
OR4C15	OR4C6	OR4M1	OR4M2	OR5D18
OR5F1	OR5L1	OR5L2	OR6F1	OR8H2
OR8I2	OR8U1	ORC4	PAK5	PALB2
PARP1	PARP2	PARP3	PARP4	PAX5
PBRM1	PCBP1	PCDH17	PDCD1	PDCD1LG2

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PDE1A	PDE1C	PDGFRA	PDGFRB	PDIA3
PGD	PHF6	PICALM	PIK3C2B	PIK3CA
PIK3CB	PIK3CD	PIK3CG	PIK3R1	PIK3R2
PIM1	PLCG1	PLXDC2	PMS1	PMS2
POLD1	POLE	POM121L12	POT1	PPARG
PPFIA2	PPM1D	PPP2R1A	PPP2R2A	PPP6C
PRDM1	PRDM9	PRKACA	PRKACB	PRKAR1A
PSMB10	PSMB8	PSMB9	PTCH1	PTEN
PTH	PTPN11	PTPRD	PTPRT	PXDNL
RAC1	RAD50	RAD51	RAD51B	RAD51C
RAD51D	RAD52	RAD54L	RAF1	RARA
RASA1	RASA2	RB1	RBM10	RBP3
RECQL4	REG1A	REG1B	REG3A	REG3G
RELA	RET	RGS7	RHEB	RHOA
RICTOR	RIT1	RNASEH2A	RNASEH2B	RNASEH2C
RNF213	RNF43	ROS1	RPA1	RPL10
RPL22	RPL5	RPS6KB1	RPTN	RPTOR
RSPO2	RSPO3	RUNDC3B	RUNX1	RUNX1T1
SDHA	SDHB	SDHC	SDHD	SETBP1
SETD2	SF3B1	SH3RF2	SIX1	SIX2
SLC15A2	SLC5A5	SLC8A1	SLCO1B3	SLX4
SMAD2	SMAD4	SMARCA4	SMARCB1	SMC1A
SMO	SNCAIP	SOCS1	SOS1	SOX2
SOX9	SPEN	SPOP	SRC	SRSF2
STAG2	STAT1	STAT3	STAT5B	STAT6
STK11	SUFU	SUZ12	SYN2	SYT10
SYT16	TAF1	TAP1	TAP2	TAPBP
TBX3	TCF7L2	TERT	TET2	TFE3
TFEB	TG	TGFBR1	TGFBR2	TMEM132D
TNFAIP3	TNFRSF14	TOP1	TOP2A	TP53
TP63	TPMT	TPP2	TPTE	TRHDE
TRIM48	TRIM51	TRRAP	TSC1	TSC2
TSHR	U2AF1	UGT1A1	USP8	USP9X
VHL	WAS	WT1	XPO1	XRCC2

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XRCC3	YAP1	YES1	ZBTB20	ZFHX3
ZIM3	ZMYM3	ZNF217	ZNF429	ZNF479
ZNF536	ZRSR2			

RNA Gene List				
ALK	MET	NTRK1	NTRK2	NTRK3
ROS1				

TMB	Category	Interpretation
1-5 Mutations/Mb	Low	Unlikely to benefit from Immuno-Oncology therapy.
6-19 Mutations/Mb	Intermediate	Moderate response anticipated towards Immuno-Oncology therapy.
>20 Mutations/Mb	High	Good and significant response expected to Immuno-Oncology therapy

MSI	% of Unstable Loci
MSI - Stable	0-19%
MSI - Low / Intermediate	20-30%
MSI - High	>20%

Tier	Interpretation
Tier IA	Variants of Strong Clinical Significance: Biomarkers that predict response or resistance to US FDA-approved therapies for a specific type of tumor or have been included in professional guidelines as therapeutic, diagnostic, and/or prognostic biomarkers for specific types of tumors.
Tier IB	Variants of Strong Clinical Significance: Biomarkers that predict response or resistance to a therapy based on well-powered studies with consensus from experts in the field, or have diagnostic and/or prognostic significance of certain diseases based on well-powered studies with expert consensus.
Tier IIC	Variants of Potential Clinical Significance: biomarkers that predict response or resistance to therapies approved by FDA or professional societies for a different tumor type (i.e., off-label use of a drug), serve as inclusion criteria for clinical trials, or have diagnostic and/or prognostic significance based on the results of multiple small studies.
Tier IID	Variants of Potential Clinical Significance: Biomarkers that show plausible therapeutic significance based on preclinical studies or may assist disease diagnosis and/or prognosis themselves or along with other biomarkers based on small studies or multiple case reports with no consensus.
Tier III	Variants of Unknown clinical significance (VUS): Not observed at a significant allele frequency in the general or specific subpopulation or pancancer or tumor specific variant databases. No convincing

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	published evidence of cancer Association

References:

1. *Exome Aggregation Consortium et al. "Analysis of Protein-Coding Genetic Variation in 60,706 Humans." Nature (2016) 536(7616): 285–291. PMID: 27535533.*

2. *Forbes, Simon A., et al. "COSMIC: somatic cancer genetics at high-resolution." Nucleic Acids Research (2016) 45(D1): D777-D783. PMID: 27899578.*

3. *Li, Marilyn M., et al. "Standards and guidelines for the interpretation and reporting of sequence variants in cancer: a joint consensus recommendation of the Association for Molecular Pathology, American Society of Clinical Oncology, and College of American Pathologists." The Journal of Molecular Diagnostics (2017) 19(1): 4-23. PMID: 27993330.*

4. *Mamanova L, Coffey AJ, Scott CE, Kozarewa I, Turner EH, Kumar A, Howard E, Shendure J, Turner DJ: Target-enrichment strategies for next-generation sequencing. Nat Methods 2010, 7: 111–118*

5. *Richards, Sue, et al. "Standards and guidelines for the interpretation of sequence variants: a joint consensus recommendation of the American College of Medical Genetics and Genomics and the Association for Molecular Pathology." Genetics in Medicine (2015) 17(5): 405. PMID: 25741868.*

6. *The 1000 Genomes Project Consortium. "A Global Reference for Human Genetic Variation." Nature (2015) 526(7571): 68–74. PMID: 26432245.*

7. *Thusberg J, Olatubosun A, Vihinen M. Performance of mutation pathogenicity prediction methods on missense variants. Hum Mutat. 2011;32:358–368. doi: 10.1002/humu.21445.*

------------------ End Of Report ------------------

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